

INVITED REVIEW



Artificial intelligence streamlines scientific discovery of drug–target interactions

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Abstract

Drug discovery is a complicated process through which new therapeutics are identified to prevent and treat specific diseases. Identification of drug–target interactions (DTIs) stands as a pivotal aspect within the realm of drug discovery and development. The traditional process of drug discovery, especially identification of DTIs, is marked by its high costs of experimental assays and low success rates. Computational methods have emerged as indispensable tools, especially those employing artificial intelligence (AI) methods, which could streamline the process, thereby reducing costs and time consumption and potentially increasing success rates. In this review, we focus on the application of AI techniques in DTI prediction. Specifically, we commence with a comprehensive overview of drug discovery and development, along with systematic prediction and validation of DTIs. We proceed to highlight the prominent databases and toolkits used in developing AI methods for DTI prediction, as well as with methodologies for evaluating their efficacy. We further extend the exploration into three primary types of state-of-the-art AI methods used in DTI prediction, including classical machine learning, deep learning and network-based methods. Finally, we summarize the key findings and outline the current challenges and future directions that AI methods face in scientific drug discovery and development.

KEYWORDS

artificial intelligence, drug-target interaction

1 | INTRODUCTION

Drug discovery remains a formidable challenge within the realm of medicine and healthcare, with the identification of drug–target interactions (DTIs) being a pivotal (Zeng et al., 2019; Zeng, Zhu,

et al., 2020b). DTIs reveal whether a drug compound can bind to target proteins and decipher how such binding interactions occur. Traditional approaches to identify DTIs heavily rely on costly and time-consuming *in vitro* and *in vivo* experiments (Paul et al., 2010). However, recent advances in computational techniques, particularly artificial intelligence (AI) methods, have made remarkable progress in this field (Zeng et al., 2019; Zeng, Wang, et al., 2022a; Zeng, Zhu, et al., 2020b). AI methods offer multiple benefits such as high prediction accuracy, reduced costs and fast prediction speeds (Ye et al., 2021).

Abbreviations: AD, Alzheimer's disease; AI, artificial intelligence; AUROC, area under the receiver operating characteristic; BERT, bidirectional encoder representations from transformers; DTIs, drug–target interactions; GAT, graph attention network; GCN, graph convolutional network; MS, multiple sclerosis; R, Pearson correlation coefficient; UniProt, Universal Protein Resource.

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Early applications of AI methods in DTI prediction predominantly leveraged classical machine learning methods, which harness manually curated descriptors or features of drugs or targets (Bagherian et al., 2021; Vamathevan et al., 2019). However, a significant challenge of these methods is their fixed-size input requirement, hampering their generalization ability to diverse datasets. In contrast, deep learning methods gained popularity because of their ability to handle larger datasets, better performance and ability to learn intricate relationship between input data and output (LeCun et al., 2015). The successes of deep learning across different domains, such as computer vision (Voulodimos et al., 2018), natural language processing (Otter et al., 2020) and reinforcement learning (Arulkumaran et al., 2017) have spurred its application in DTI prediction.

The past decade has witnessed a surge in AI-driven DTI prediction research, facilitated by the widespread availability of data repositories housing compounds, substances and protein data, improved computational power and increasingly sophisticated AI algorithms. Even though there have been several review articles focusing on AI methods for computational drug discovery, they often emphasize specific approaches, such as deep learning approaches (Abbasi et al., 2021; Askr et al., 2023), feature-based machine learning approaches (Sachdev & Gupta, 2019), molecular similarity-based machine learning approaches (Ding et al., 2014) and network-based approaches (Liu et al., 2020; Zeng, Zhu, et al., 2020b). Yet there remains a pressing need for a comprehensive overview encompassing all AI techniques in DTI prediction. To address this gap, this review aims to provide an integrative overview of AI techniques in DTI predictions. Specifically, we begin by summarizing commonly used databases, tools, and evaluation metrics for DTI prediction and cross-validation using AI techniques. Subsequently, we provide an overview of AI techniques, especially classical machine learning methods, deep learning approaches and their applications in DTI prediction. Additionally, we highlight the application of AI techniques in drug repurposing, a specialized case of DTI prediction. Finally, we conclude this review by discussing the challenges and future directions that may impede the advancement of AI techniques for prediction of DTI.

2 | DATABASES, TOOLS AND EVALUATION METRICS

2.1 | AI-ready open data

Several databases provide essential data for the prediction of DTIs, encompassing ligand compound information, target protein information and interaction information. These databases can typically be categorized into three classes:- drug-focused databases, target-focused databases and binding databases. Notable examples of drug-focused databases include DrugBank (Wishart et al., 2018) and ChEMBL (Gaulton et al., 2012). The latest release of DrugBank Online (Version 5.1.12 released on 14 March 2024) contains 17,085 drugs, with 2786 approved small-molecule drugs, 1629 approved biologics and many other molecules under clinical or experimental investigations. ChEMBL stores 2D structures, a variety of properties and bioactivities for more

than 2.4 million bioactive drug-like compounds (ChEMBL34 updated on 28 March 2024). Examples of target-focused databases include Universal Protein Resource (UniProt) (UniProt Consortium, 2019), Protein Data Bank (PDB) (Berman et al., 2000) and AlphaFold Protein Structure Database (Varadi et al., 2024). UniProt stores sequences, functions and properties for more than 245 million proteins (accessed July 2024). However, only less than 600,000 proteins have been manually annotated and reviewed by experts. The human proteome is estimated to contain more than 82,000 proteins (UniProt Proteome: UP000005640). Protein Data Bank contains more than 200,000 3D protein structure data, which are determined through experiments such as X-ray crystallography and nuclear magnetic resonance (NMR) spectroscopy. Such experimental methods can deliver highly accurate protein structure data, but they are characterized by high costs and slow progression. The recent progression of computational methods, especially AI methods, for protein structure predictions has revolutionized the landscape of protein structure discovery. AlphaFold Protein Structure Database holds more than 214 million protein structures predicted by AlphaFold2 (Jumper et al., 2021). BindingDB (Gilson et al., 2016) and PDBbind (Wang et al., 2005) are databases that catalogue information about the binding affinity between drugs and targets. BindingDB contains more than 2.9 million measurements with more than 1.3 million compounds and more than 9000 targets. A comprehensive overview of commonly used databases for identifying DTI is provided in Table 1.

2.2 | Tools for molecular representation

Several tools are available for computing the physiochemical features of both drugs and targets. Here, we list and summarize commonly used toolkits into three categories:- tools for computing feature vectors, tools for computing similarities and tools for generating structures.

2.2.1 | Tools for computing feature vectors

1. PaDEL (Yap, 2011): PaDEL is a software that calculates descriptors and fingerprints for molecules. PaDEL can compute 1875 different 1D, 2D and 3D descriptors and 12 types of fingerprints. PaDEL can take both molecular structure file and SMILES string as input.
2. ProtDCAI (Ruiz-Blanco et al., 2015): ProtDCAI is a software to compute features for proteins based on 3D structures and amino acid sequences. ProtDCAI can generate 1998 features where 999 features are from sequence and 999 features are from structure.
3. ProtParam (Gasteiger et al., 2005): ProtParam is a web server that can compute numerous features of proteins, including molecular weight, atomic composition and amino acid composition. ProtParam only takes amino acid sequences as input.
4. PubChemPy (Swain, 2014): PubChemPy is a Python toolkit that provides users with the ability to calculate chemical fingerprints. All computational work is performed on PubChem's servers through PubChem's database and toolkits.

TABLE 1 Commonly used databases in drug–target interaction prediction.

Database name	Description	Link
Drug-focused databases		
DrugBank (Wishart et al., 2018)	A database of drugs and drug targets with chemical, pharmaceutical, sequence, structure and other information.	https://go.drugbank.com/
ChEMBL (Gaulton et al., 2012)	A manually curated database of drug-like bioactive compounds with binding, functional and other information.	https://www.ebi.ac.uk/chembl/
ZINC20 (Irwin et al., 2020)	A database of commercially available and annotated compounds.	https://zinc20.docking.org/
PubChem (Kim et al., 2023)	A database of molecules incorporating various molecule properties, features, structures and other information.	https://pubchem.ncbi.nlm.nih.gov/
DrugCentral (Ursu et al., 2016)	A database of drugs with description of usage, dosage, ADMET properties and other information.	https://drugcentral.org/
SIDER (Kuhn et al., 2010)	A database that contains side effect and adverse drug reactions of marketed medicines.	http://sideeffects.embl.de/
Target-focused databases		
UniProt (UniProt Consortium, 2019)	A database of protein sequences, structures and functional information.	https://www.uniprot.org/
Protein Data Bank (Berman et al., 2000)	A database of protein's 3D structure determined by experimentations.	https://www.rcsb.org/
AlphaFold Protein Structure Database (Jumper et al., 2021)	A database hosting protein structures predicted by AlphaFold2.	https://alphafold.ebi.ac.uk/
Binding interaction databases		
BindingDB (Gilson et al., 2016)	A database that includes binding affinities of proteins with drug-like small molecules.	http://www.bindingdb.org/
PDBbind (Wang et al., 2005)	A database of binding affinity data of compound–protein interactions measured using experiments.	http://www.pdbbind.org.cn/

2.2.2 | Tools for computing molecular similarities

1. SIMilar COMpound (SIMCOMP) (Hattori et al., 2010): SIMCOMP is a graph-based method to compare chemical structures. Each graph represents a chemical compound with nodes representing atoms and edges representing covalent bonds. Drug similarity is computed based on the maximal common substructure between the two drug structures.
2. Compound–protein interaction with R programming language (Rcpi) (Cao et al., 2015): Rcpi is a package available in R programming language that offers several functions including similarity computation. Similarities between proteins are derived from protein sequence alignment and Gene Ontology (GO) semantic similarity. Similarities between compounds are based on molecular fingerprints and maximal common substructure.

2.2.3 | Tools for generating chemical structures

1. Open Babel (O'Boyle et al., 2011): Open Babel is a software that encompasses several important features for drug discovery, especially generating 2D and 3D structures of molecules from Simplified Molecular Input Line Entry System (SMILES) strings.
2. RDKit (Landrum, 2013): RDKit is a collection of cheminformatics tools with one important feature of generating 2D and 3D structures of molecules from SMILES strings. RDKit is

available in multiple programming languages, including C++ and Python.

While some cheminformatics toolkits and software offer singular functionalities, others provide a comprehensive suite of features. For example, RDKit goes beyond generating 2D and 3D molecular structures. RDKit (Landrum, 2013) can also calculate feature vectors for molecules and perform similarity computations. This versatility of RDKit, coupled with its user-friendly Python interface (the most popular language for building AI models), makes RDKit a popular choice for researchers in drug discovery. Researchers can leverage RDKit's diverse functionalities to streamline different tasks within the AI-based drug discovery frameworks.

2.3 | Metrics

Evaluation metrics provide a quantitative means to measure the performance of AI-driven DTI prediction methods, categorized into classification and regression tasks, each utilizing different evaluation metrics. Commonly used evaluation metrics for classification tasks include the following:

1. Positives and negatives: True positive (TP), true negative (TN), false positive (FP) and false negative (FN) describe correct and incorrect predictions based on positive and negative labels.

2. Accuracy: Accuracy measures the proportion of correct predictions over the entire dataset:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

3. Precision: Precision measures the fraction of correctly predicted positive results among all predicted positive results:

$$\text{Precision} = \frac{TP}{TP + FP}$$

4. Recall: Recall measures the fraction of correctly predicted positive samples over all actual positive samples:

$$\text{Recall} = \frac{TP}{TP + FN}$$

5. F1-score: F1-score is the harmonic mean of the precision and recall:

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN}$$

6. Receiver operating characteristic (ROC) curve: The ROC curve illustrates the TP (true positive) rate against FP (the false positive) rate at various discrimination thresholds.

When the dataset is balanced across different classes, accuracy and area under the ROC (AUROC) curve are good choices to evaluate the model's performance. However, when the dataset is imbalanced, accuracy and the AUROC curve can be misleading. For example, if a dataset contains more negative samples than positive samples and a model is only good at predicting negatives and not good at predicting positives, the AUROC curve can still be high. In this case, using F1-score is a better choice as F1-score can provide a fairer evaluation of the model (Jeni et al., 2013).

Commonly used evaluation metrics for regression tasks include the following:

1. Mean absolute error (MAE): MAE assesses the absolute error between the predicted and actual values:

$$\text{MAE} = \frac{\sum_{i=1}^n |x_i - y_i|}{n}$$

2. Root mean squared error (RMSE): RMSE assesses the difference between the predicted and actual values, accounting for their squared differences:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (x_i - y_i)^2}{n}}$$

3. Mean squared logarithmic error (MSLE): MSLE assesses squared logarithmic error between the predicted and actual values:

$$\text{MSLE} = \frac{\sum_{i=1}^n (\log(x_i + 1) - \log(y_i + 1))^2}{n}$$

4. Mean absolute percentage error (MAPE): MAPE assesses the prediction accuracy of a regression model without being affected by target variable scaling:

$$\text{MAPE} = 100 \times \frac{\sum_{i=1}^n \frac{|x_i - y_i|}{x_i}}{n}$$

5. Pearson correlation coefficient (R): The Pearson R correlation coefficient assesses the linear correlation between the predicted and actual values:

$$R = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

In the evaluation metrics described above, x_i represents the actual (observed) values, while y_i represents the predicted values for each data point i in the dataset, and $\bar{x}$ and $\bar{y}$ represent the mean of the actual values and predicted values. Compared to root mean squared error (RMSE), mean absolute error is less sensitive to outliers. This is because outliers are squared in the calculation of root mean squared error, leading to a disproportionate influence on the final metric. Therefore, mean absolute error is a better choice when dealing with datasets containing a significant number of outliers. When error scale should match target scale, root mean squared error is a better choice as root mean squared error is more scale-dependent. Mean absolute percentage error expresses the error as a percentage of the actual difference, making it scale-independent metric. However, mean absolute percentage error can be problematic for datasets containing zero values in the actual target variable, as division by zero is undefined. Additionally, both mean absolute error (MAE) and mean absolute percentage error (MAPE) are generally less computationally expensive to calculate compared to other evaluation metrics. Ultimately, the best choice of evaluation metric depends on the specific characteristics of the DTI prediction task and dataset.

3 | AI-POWERED PREDICTION OF DRUG-TARGET INTERACTION (DTI)

Within the realm of DTI predictions, commonly used AI techniques encompass six major types of models:- feature vector-based models, similarity-based models, sequence-based models, network-based

models, image-based models and structure-based models (Zeng, Wang, et al., 2022a). Applying AI models to predict DTIs comprises three fundamental steps: dataset processing, model training and making predictions. Dataset processing involves the extraction of important features from input drug and target data or the direct utilization of raw data. Subsequently, the dataset is split into three subsets: the training set, the validation set and the test set, wherein the training set is used to

facilitate the AI model's learning process, the validation set is used to fine-tune hyperparameters of the AI model and the test set assesses the AI model's performance on unseen data (Figure 1). Following this, the trained AI model is applied to the test set of DTI data and, possibly, additional data that are not observed during the training process.

The AI model's architecture and training approach must be tailored to the specific DTI prediction task and dataset available. For classification tasks, such as determining whether a drug interacts with a given target, the model should output binary predictions. In contrast, regression tasks, like estimating the binding affinity between drugs and targets, require the model to produce continuous numerical outputs. This fundamental distinction in the prediction objective necessitates appropriate adjustments to the model's structure, loss function and evaluation metrics to ensure optimal performance for the intended DTI prediction scenario. In the following sections, we delve into the intricateness of AI models used for prediction of DTIs and provide a brief description, advantage and disadvantage of these AI methods in Table 2.

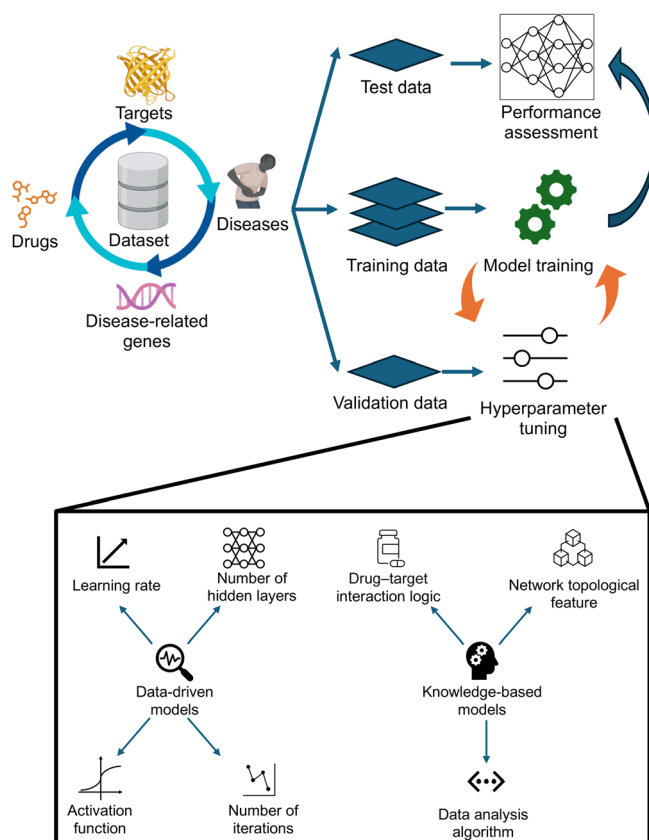


FIGURE 1 Pipeline of applying artificial intelligence (AI) techniques for drug discovery. The entire pipeline consists of several key steps. Data acquisition and preprocessing involve the collections of datasets for drugs, targets and potentially diseases, and disease-related genes. The data are then processed and split into three subsets: a training dataset to train AI models; a validation dataset to help tune hyperparameters; and a test dataset to assess the performance of AI models. Training of AI models is to find the optimal parameters of the model based on the training dataset. Hyperparameter tuning is to find the best set of hyperparameters of AI models so that AI models can have the best performance using existing training data. Tuning of hyperparameters for data-driven AI models, such as classical machine learning or deep learning models, mainly consists of choosing optimal model architecture (like number of hidden layers and activation function) and better optimization strategy, including learning rate and number of training iterations. However, knowledge-based AI models, such as network-based models, require selecting optimal knowledge-related hyperparameters, such as drug–target interaction logic and topological features of drug–target interaction networks. Testing of AI models using data not available in the training dataset can confirm the AI model's generalization ability.

3.1 | Fingerprint-based models

Fingerprint-based models begin by calculating physiochemical fingerprints for drugs and target proteins (Figure 2a, left). These fingerprints capture relevant chemical features, often represented using circular fingerprints (Glen et al., 2006), Molecular ACCESS System (MACCS) fingerprints (Cereto-Massagué et al., 2015; Durant et al., 2002) and PubChem fingerprints (Fernández-de Gortari et al., 2017). To improve computational efficiency for classical machine learning models and potentially reduce redundant features, dimensionality reduction techniques can be employed. Principal component analysis (PCA) (Wold et al., 1987) is a popular choice alongside techniques like t-distributed stochastic neighbour embedding (t-SNE) (Hinton & Roweis, 2002), non-negative matrix factorization (NMF) (Sra & Dhillon, 2005), singular value decomposition (SVD) (Golub & Reinsch, 1971) and Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018). Principal component analysis and UMAP tend to capture global features of the whole dataset, while t-distributed stochastic neighbour embedding (t-SNE) tends to maintain the local relationship between data samples. Following feature computation and potential dimensionality reduction, different classical machine learning models are employed to learn from these features and predict DTIs. These models can be categorized as classification models like support vector machine (SVM) (Hearst et al., 1998) and random forest (RF) (Breiman, 2001), or regression models like Bayesian regression (Bishop & Tipping, 2003) or logistic regression (Kleinbaum et al., 2002). Early DTI prediction studies using feature-based approaches often relied on single-model architectures. Examples include support vector machine-based models by Wang et al. (2010), RF-based models by Ahn et al. (2022) and Shi et al. (2019), and rotation forest models by Zhan et al. (2020). These single-model approaches, however, have limited ability to achieve better performance and generalizability. Recent advancements have seen a shift

TABLE 2 Recently developed artificial intelligence methods for prediction of drug–target interaction.

Model type	Model	Description	Year
Classical machine learning models	Feature vector-based models		
	Wang et al. (Wang et al., 2010)	An SVM model that used features extracted from drug–protein pairs to predict DTIs.	2010
	Ahn et al. (Ahn et al., 2022)	An RF model that used KLD to calculate drug features to predict DTIs.	2022
	Shi et al. (Shi et al., 2019)	An ensemble model that uses LASSO and RF to predict DTIs. Drug–target features are extracted using PsePSSM and FP2 molecular fingerprinting.	2019
	Zhan et al. (Zhan et al., 2020)	An RF model that uses drug features calculated using LOOP and PSSM. This model also achieves high performance on enzyme, ion channel, GPCRs and nuclear receptor data beyond benchmark data.	2020
	NegStacking (Yang et al., 2020)	An ensemble model that integrates decision trees and logistic regression to predict DTI. NegStacking innovatively leveraged a large number of negative samples in the dataset to improve the performance.	2020
	Ezzat et al. (Ezzat et al., 2017)	An ensemble model that integrates dimensionality reduction, decision trees and kernel ridge regressors to predict DTI. Features used in this work included constitutional, topological and geometrical descriptors for drugs and amino acid composition, pseudo-amino acid composition, and CTD descriptors for target proteins.	2017
	Similarity-based models		
	Bleakley and Yamanishi (Bleakley & Yamanishi, 2009)	A similarity-based model based on a bipartite local model. DTIs are represented in bipartite graphs. This model can both predict target protein given a drug and predict potential drug given a target protein.	2009
	DrugE-Rank (Yuan et al., 2016)	A similarity-based model that borrows idea from ensemble learning. DrugE-Rank consisted of six machine learning components. DrugE-Rank was thoroughly validated using FDA-approved drugs and FDA experimental drugs.	2016
	Jacob and Vert (Jacob & Vert, 2008)	A similarity-based model that predicts DTI for targets with no known 3D structure and few or no known interacting drugs. This model was able to discover potential drugs for enzymes, GPCRs and ion channels.	2008
	Xia et al. (Xia et al., 2010)	A similarity-based model that was trained using semi-supervised learning manner. This model integrated DTI information with chemical structures and genomic sequence data.	2010
	Gönen (Gönen, 2012)	A similarity-based model that combines Bayesian formulation, dimensionality reduction, matrix factorization and binary classification. This model uses chemical similarity between drugs and genomic similarity between targets.	2012
Deep learning models	Sequence-based models		
	WideDTA (Öztürk et al., 2019)	A CNN model that used SMILES string and amino acid sequence as input. Beyond using sequences, WideDTA also encodes ligand maximum common substructures and protein domains and motifs into the model to improve performance.	2019
	SMILES2Vec (Goh, Hodas, et al., 2017a)	An RNN model that directly takes SMILES string as input. The authors also improved the interpretability of the model by visualizing the attention of the model on the molecule.	2017
	DeepSMILES (O'Boyle & Dalke, 2018)	A transformer model that takes SMILES as input. DeepSMILES addressed the potential syntax problems in the SMILES by eliminating unmatched ring closure and adopting a postfix notation.	2018
	AttentionDTA (Zhao et al., 2019)	A sequence-based model that uses CNN to extract representations from SMILES strings and amino acid sequences and a transformer model to combine the representations and predict DTIs.	2019
	MT-DTI (Shin et al., 2019)	A sequence-based model that uses transformer to extract representations from SMILES strings and CNN to extract representations from amino acid sequences.	2019
	SMILES Transformer (Honda et al., 2019)	A transformer model in an encoder–decoder structure. SMILES Transformer adopted a pretraining strategy where the model reconstructs the input SMILES string, to improve the performance of the model.	2019
	SMILES-BERT (Wang et al., 2019)	A transformer model with only an encoder. SMILES-BERT was pretrained to predict the masked part of the input SMILES string.	2019

TABLE 2 (Continued)

Model type	Model	Description	Year
	CHEM-BERT (Kim et al., 2021)	A transformer encoder model. CHEM-BERT uses a similar pretraining strategy as SMILES-BERT. CHEM-BERT offers a web server where users can submit job to predict DTIs and drug properties online.	
	Image-based models		
	Chemception (Goh, Siegel, et al., 2017b)	A CNN model that just uses 2D snapshots of molecules as input. Chemception was trained on a dataset with less than 40,000 compounds to predict different properties and DTIs.	2017
	DEEPScreen (Rifaiglu et al., 2020)	A CNN model that focuses on 2D structure representations of compounds. DEEPScreen was trained to predict DTIs for 704 target proteins.	2020
	ImageMol (Zeng, Xiang, et al., 2022b)	A CNN model that adopts a pretraining strategy to learn important chemical information from molecules. ImageMol was used to discover potential treatments to COVID-19.	2022
	Structure-based models		
	MolCLR (Wang, Wang, et al., 2022b)	A GCN model that takes molecular structures as input. MolCLR was pretrained on around 10 million molecules to improve the generalization ability of the model.	2022
	N-gram graph (Liu et al., 2019)	An unsupervised GNN model that constructs representations for molecular structures based on covalent bonds. The N-gram graph was rigorously assessed on 60 different tasks.	2019
	MGCN (Lu et al., 2019)	A GCN model that studied conformation and spatial information using molecular graphs. MGCN was designed in a hierarchical manner to maximize the effectiveness of the model.	2019
	DGraphDTA (Jiang et al., 2020)	A GNN model that focuses on proteins and constructs protein graphs based on protein contact maps. DGraphDTA showed strong generalization ability and robustness across multiple benchmark datasets.	2020
Network-based models	Network-based models		
	GCN-DTI (Zhao et al., 2021)	A GCN model that uses drug-protein pairs network as input. The drug-protein pair network used in GCN-DTI involves three types of drug-protein associations: strong, weak and non-association.	2021
	DTI-GAT (Wang et al., 2021)	An end-to-end GAT model that processes drug-protein interaction network with attention mechanism. DTI-GAT harnesses interaction patterns and features from drug and protein sequences.	2021
	DTI-HETA (Shao et al., 2022)	An end-to-end network-based model that uses both GCN and GAT. GCN studies pattern from heterogeneous graph from the source dataset, while GAT highlights the contributions of neighbouring nodes.	2022
	DTI-GTN (Wang, Guo, et al., 2022a)	A GTN model that innovatively considered each drug-protein interaction as a node. PCA was used in this work to reduce the dimensionality of feature vectors of drugs and proteins.	2022
	AI-Bind (Chatterjee et al., 2023)	A network-based framework that leverages unsupervised pretraining improving the generalization ability of DTI predictions. Experimental validations are conducted to support the prediction results.	2023
	deepDTnet (Zeng, Zhu, et al., 2020b)	An encoder-decoder model that takes graph representations as input. deepDTnet embeds 15 different interaction networks, including chemical, genomic, phenotypic and cellular networks.	2020
	deepDR (Zeng et al., 2019)	A multimodal deep autoencoder model that learns high-level features of drugs from heterogeneous interaction networks. deepDR integrates 10 different types of interaction networks, including drug-disease, drug-side effect, drug-target and drug-drug networks.	2019

Abbreviations: CNN, convolutional neural network; COVID-19, coronavirus disease 2019; CTD, composition, transition and distribution; DTI, drug-target interaction; FDA, Food and Drug Administration; GAT, graph attention network; GCN, graph convolutional network; GNN, graph neural network; GPCRs, G protein-coupled receptors; GTN, graph transformer network; KLD, Kullback-Leibler divergence; LASSO, least absolute shrinkage and selection operator; LOOP, local optimal-oriented pattern; PCA, principal component analysis; PsePSSM, pseudo-position-specific scoring matrix; PSSM, position-specific scoring matrix; RF, random forest; RNN, recurrent neural network; SMILES, Simplified Molecular Input Line Entry System; SVM, support vector machine.

towards ensemble models that combine several classical machine learning models. This approach aims to improve prediction performance and model robustness and reduce the risk of overfitting and underfitting (Dong et al., 2020). Examples of ensemble models for DTI prediction include NegStacking by Yang et al. (2020) and the ensemble approach by Ezzat et al. (2017). These feature vector-based AI models can be efficient when the dataset is small and well structured, especially for chemical fingerprints, which are a well-structured data type. However, a key challenge lies in the strategy of feature construction and selection. For example, a larger feature set may include redundant features that can lead to higher computational cost and overfitting issue, while a smaller feature set may not contain enough features for the model to learn from leading to underfitting issue.

3.2 | Similarity-based models

Similarity-based models assume drugs with high similarity in chemical structure, side effect profile or gene expression response are likely to interact with similar targets and vice versa for targets with high similarity in sequence, protein–protein interaction network proximity or gene ontology (Mitchell, 2001). These methods first calculate two similarity matrices. One matrix captures the pairwise similarities between drugs and the other matrix captures similarities between targets. Once the similarity matrices are generated, different classical machine learning algorithms can be employed to predict DTIs (Figure 2a, right). Commonly used classical machine learning algorithms include *k*-nearest neighbour (kNN) as a baseline prediction or as a part of ensemble models (Bleakley & Yamanishi, 2009; Yuan et al., 2016). Other algorithms like support vector machine can be used to identify non-linear relationships between similarities and interactions (Jacob & Vert, 2008). Additionally, bipartite local model (BLM) has been employed to exploit bipartite properties within the similarity matrices (Bleakley et al., 2007; Bleakley & Yamanishi, 2009; Mordelet & Vert, 2008). Regularized least squares (RLS) (Xia et al., 2010) and kernelized Bayesian matrix factorization (Gönen, 2012) are additional examples of classical machine learning approaches to predict DTIs based on similarities. Compared to feature vector-based models, these similarity-based models are more efficient, as feature construction is not required for similarity-based models.

Even though similarity-based models show promising results in DTI prediction, one major limitation of these models is the assumption that similar drugs share similar targets and vice versa. This assumption limits the performance of similarity-based models on some DTI prediction tasks when similarity between drugs or targets is not high enough (Ding et al., 2014).

3.3 | Sequence-based models

Sequence-based models are inspired by natural language processing (NLP) techniques. Drugs are typically represented by the Simplified Molecular Input Line Entry System (SMILES) (Weininger, 1988) strings, while proteins are represented by their amino acid

sequences (Figure 2b). Early works utilized recurrent neural networks (RNNs) (Medsker & Jain, 1999) or long short-term memory (LSTM) (Hochreiter & Schmidhuber, 1997), such as WideDTA (Öztürk et al., 2019), SMILES2Vec (Goh, Hodas, et al., 2017a) and DeepSMILES (O'Boyle & Dalke, 2018). These models achieved promising results, but they were limited by their inherent sequential processing nature. This sequential processing property restricts their ability to handle very long SMILES strings or protein sequences efficiently, hindering their scalability for complex drugs and targets. Attention technique and transformer model (Vaswani, 2017) have been proposed to overcome the limitations of recurrent neural networks and long short-term memories. The attention technique allows the model to focus on relevant parts of the sequence during processing, significantly improving the ability to capture long-range dependencies within the data. Transformer models, based on attention technique, addressed the efficiency bottlenecks of recurrent neural networks and long short-term memories by increasing parallelization capabilities. These advancements led to the development of various sequence-based models leveraging attention and transformer techniques, such as AttentionDTA (Zhao et al., 2019), MT-DTI (Shin et al., 2019), SMILES Transformer (Honda et al., 2019), CHEM-BERT (Kim et al., 2021) and SMILES-BERT (Wang et al., 2019). These transformer-based models have demonstrated better performance compared to recurrent neural network or long short-term memory (LSTM)-based models. Sequence-based models offer two key advantages. First, preparing the dataset is significantly easier. Unlike feature vector-based models or similarity-based models, there is no need to construct complex feature sets or similarity matrices. Second, sequence data (like SMILES strings or amino acid sequences) are inherently 1D, requiring less computational resources compared to 3D structural data. However, it is crucial to acknowledge that sequence-based models are also limited. Sequence-based models struggle to capture important information encoded in 3D structures, such as stereochemistry, which can significantly impact how a molecule interacts with its targets.

3.4 | Network-based approaches

Network-based models use connections between drugs, targets and even patients to make DTI predictions (Figure 2c). Each node in the network represents a drug, a target, a disease-related gene or a disease. Edges encode the relationship between drugs, targets, genes and diseases. Notable examples include GCN-DTI (Zhao et al., 2021) based on graph convolutional networks (GCNs) (Kipf & Welling, 2016), DTI-GAT (Wang et al., 2021) based on graph attention networks (GATs) (Velickovic et al., 2017), DTI-HETA (Shao et al., 2022) combining both GCN and GAT, DTI-GTN (Wang, Guo, et al., 2022a) based on graph transformer networks (GTNs) (Yun et al., 2019), deepDTnet (Zeng, Zhu, et al., 2020b), along with deepDR (Zeng et al., 2019). AI-Bind is another novel network-based framework that can be used to predict protein–ligand binding (Chatterjee et al., 2023). The results from AI-Bind have been validated through docking simulations and experimental validation. Network-based models excel at discovering hidden relationships between drugs and

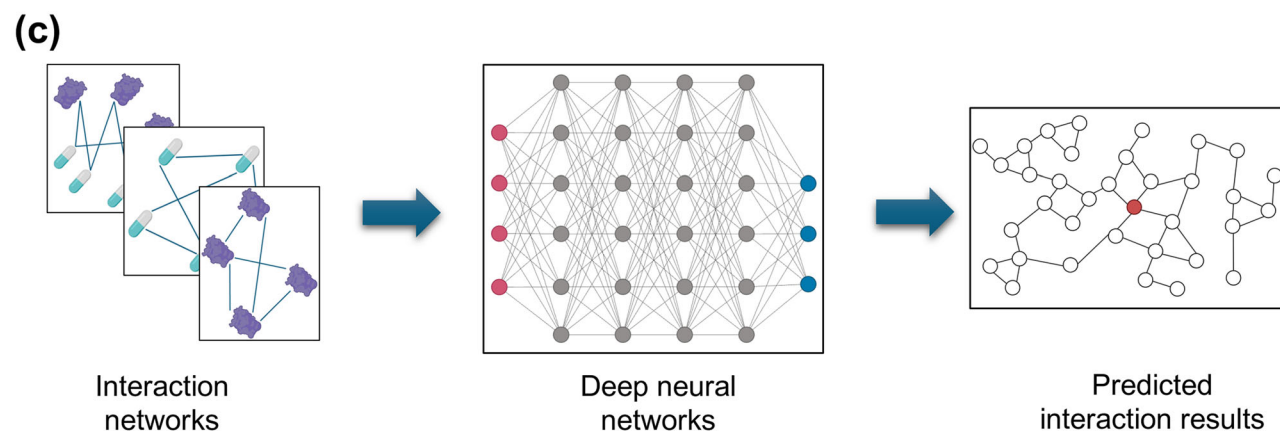
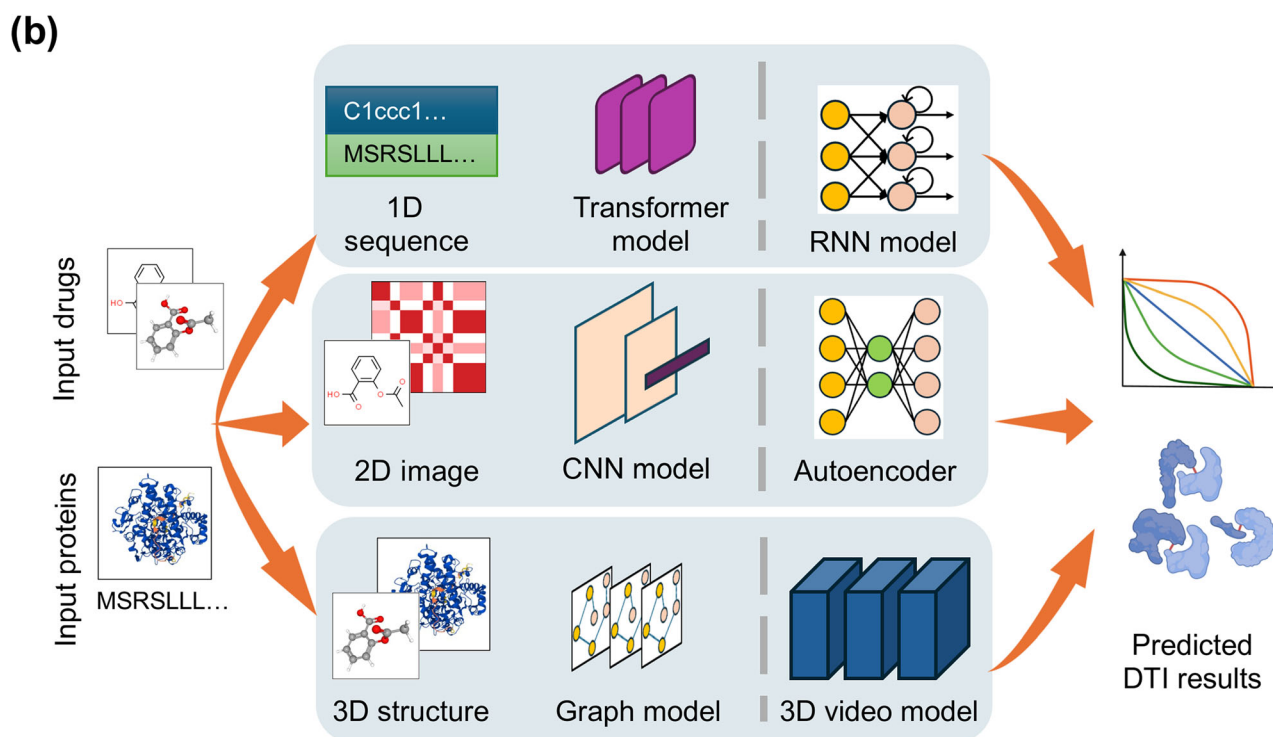
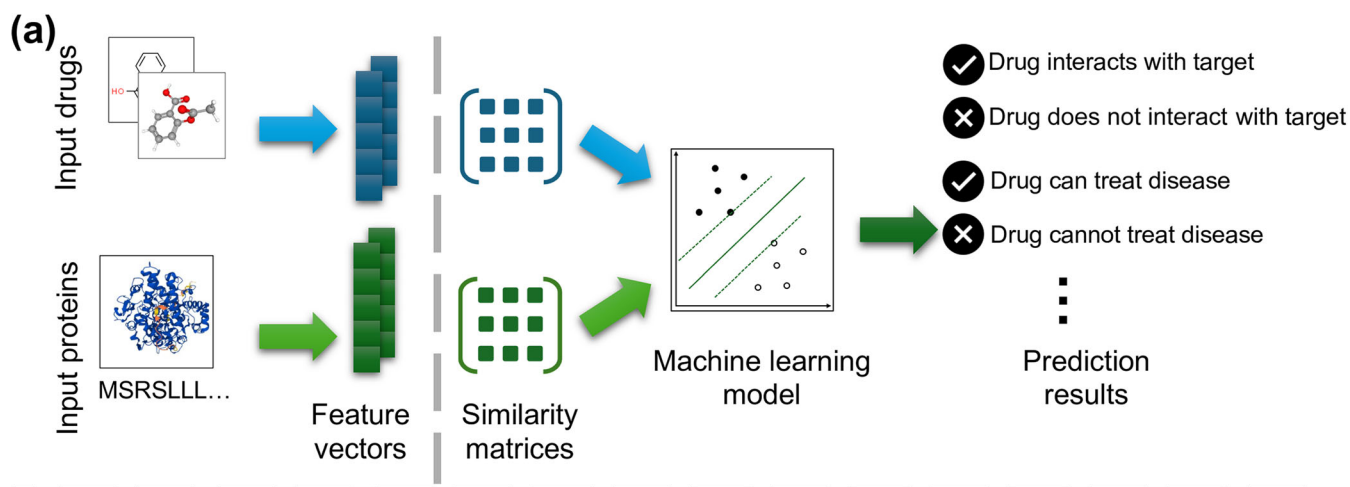


FIGURE 2 Legend on next page.

their potential targets. This advantage makes network-based models particularly well-suited for drug repurposing, where existing drugs are investigated for new targets and diseases. For example, researchers have successfully leveraged network-based models to identify potential repurposing drugs for coronavirus disease 2019 (COVID-19) (Zhou, Hou, Shen, Huang, et al., 2020a), atrial fibrillation (Lal et al., 2022), cancers (Cheng et al., 2017) and pain (Qiu et al., 2024). However, as these networks grow to encompass millions of drugs and targets, the computational complexity can become significant.

3.5 | Image-based models

Beyond sequence-based models, image-based approaches emerge as another powerful tool for DTI prediction. These models are inspired by computer vision techniques, incorporating 2D snapshot images of drug molecules as input data (Figure 2b). Compared to 1D sequence data, like SMILES strings, these images can potentially encode more information about the drug's chemical structure and spatial arrangement of atoms, which can be critical for understanding how it interacts with a target. The major backbone of image-based models is convolutional neural networks (CNNs) (Krizhevsky et al., 2012). Convolutional neural networks excel at extracting features from spatial data, making them a good candidate to analyse the 2D representations of drug molecules. Pioneering works include DEEPScreen (Rifaioğlu et al., 2020), Chemception (Goh, Siegel, et al., 2017b) and DrugCNN (Hu et al., 2019), which achieved promising results using convolutional neural networks for DTI prediction with molecular images as input. Compared to fingerprint-based models, similarity-based models, network-based models and sequence-based models, which only capture limited information from input drugs, these image-based models can capture 2D structural information from molecular images, which provides relationship between atoms, such as type of covalent bonds and bond angles, and thus leading to better performance. More recently, researchers are exploring the power of pre-training deep learning models to improve their generalization abilities. Zeng, Xiang, et al. (2022b) introduced ImageMol, a pretrained image-based deep learning model using millions of molecular images. This pretraining step allows ImageMol to effectively learn how to capture important chemical information from the input molecular image, potentially leading to more accurate predictions of drug-target interactions. LISA-CPI, (Yang et al. 2024), integrated pretrained ImageMol as well as a pretrained module in AlphaFold2 to predict DTIs. Recent advancement of AlphaFold 3 (Abramson et al., 2024) in protein

structure and biomolecular interaction predictions provides promising potential to achieve even better accuracy of protein structures as well as better performance.

3.6 | Structure-based models

Structure-based models use the 3D structures of drugs and targets for DTI prediction (Figure 2c). A popular strategy to model the 3D structures of drugs and targets is using graphs, where each node represents an atom or an amino acid and each edge represents a covalent bond or peptide bond. MolCLR (Wang, Wang, et al., 2022b) based on GCN and graph isomorphism network (GIN) (Xu et al., 2018), N-gram graph (Liu et al., 2019) and MGCN (Lu et al., 2019) based on GCN are examples of graph models that solely use drug structure data. Models like DGraphDTA (Jiang et al., 2020) and WGNN-DTA (Jiang et al., 2022) incorporate graphs of both drug structures and target structures. Compared to methods like models using fingerprints, 1D sequences or 2D images, graph models can capture more comprehensive information about the overall structure of drugs and targets. Graph models achieve this by explicitly modelling the relationship between atoms within a drug or amino acids within a target. This allows graph models to have a better understanding of how drugs and targets may interact. Graph models often exhibit lower computational complexity, making them a more favourable choice in scenarios with limited computational capabilities. However, graphs do not capture the full picture of structures of drugs and targets, as several structural information, like bond angle and lengths, remain unavailable within graph data. A more recent approach, VideoMol, leveraged video data to capture the dynamic conformation of molecules (Cheng et al., 2024). VideoMol goes beyond static graphs by considering molecular conformation at different angles and conditions. VideoMol achieved state-of-the-art performance in numerous drug discovery tasks, highlighting the potential of using video data to capture dynamic structure data for improving DTI prediction.

4 | CASE STUDIES: AI REVEALS POTENTIAL TREATMENTS FOR HUMAN COMPLEX DISEASE

Due to the slow progression of the traditional drug discovery process, AI methods have emerged as a promising solution to accelerate the discovery of potential treatments for various diseases, including infectious diseases, rare diseases and neurodegenerative diseases like

FIGURE 2 Schematic illustration of building different types of artificial intelligence (AI) models for drug-target interaction (DTI) predictions. (a) Feature vector-based and similarity-based approaches: Feature vectors or similarity matrices for drugs and targets are generated and followed by training a machine learning model on feature vectors or similarity matrices to predict DTIs. (b) Sequence-based, image-based and structure-based approaches: Drugs and targets are first described in one-dimensional (1D) sequence (SMILES string for drugs or amino acid sequence for protein), two-dimensional (2D) image snapshots or three-dimensional (3D) structures. Subsequently, deep learning models are trained on these data of sequences, images or structures and make predictions. (c) Network-based approach: Different interaction networks, such as drug-target, drug-disease and target-gene networks, are constructed. Deep neural networks are then trained on these interaction networks to discover novel treatments. CNN, convolutional neural network; RNN, recurrent neural network.

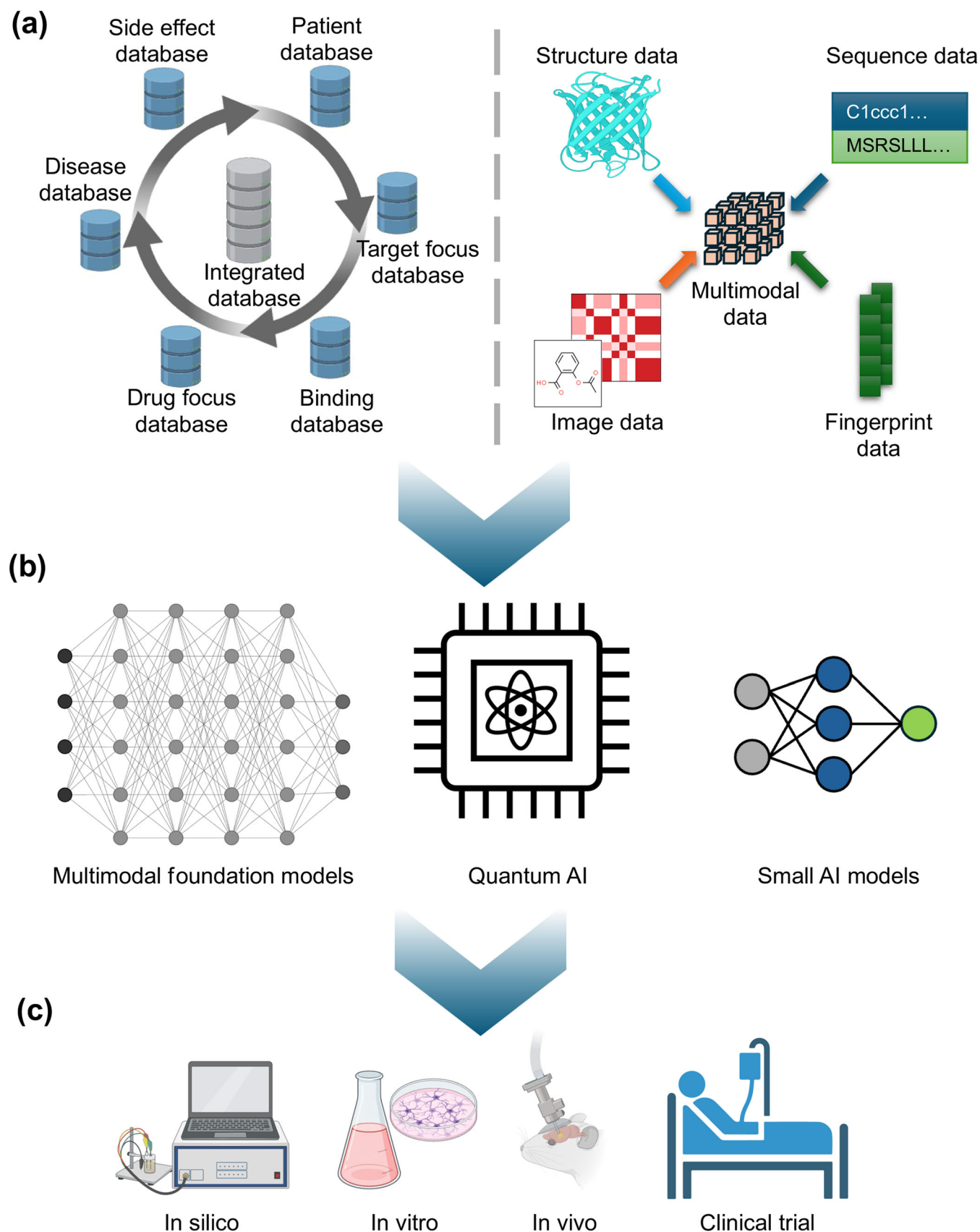


FIGURE 3 Legend on next page.

Alzheimer's disease (AD) and multiple sclerosis (MS). In this review, we use AD, MS, cardiovascular disease (CVD) and COVID-19, which are human complex diseases, as examples to study how AI techniques are being used to discover potential therapeutics to prevent and treat human complex diseases.

4.1 | Alzheimer's disease (AD)

AD, the leading cause of dementia, is characterized by progressive memory loss and cognitive decline (Blennow et al., 2006). AI techniques have shown substantial progress in drug development for AD. Classification AI models determine whether a drug interacts with AD-associated targets, while regression models quantify the strength of these interactions by predicting binding activity levels between drugs and AD-associated genes or targets. This dual approach enables researchers to both identify potential drug candidates and assess their likely efficacy, thereby streamlining the drug discovery process for AD therapeutics. One popular and well-established strategy employs network-based models to identify novel drugs, targets and genes to treat AD. For instance, a study employed a single-cell network-based machine learning framework to study the cell-specific risk genes potentially involved in AD (Gupta et al., 2022). The authors discovered novel AD-related genes using the AI model based on known AD-related genes. Another study used a network-based machine learning framework with multi-omics human protein–protein interactome data to find potential AD treatments (Fang et al., 2022). The authors not only highlighted repurposable drugs targeting AD but also investigated these repurposable drugs' mechanism of action using *in vitro* experiments in human microglia cells. Beyond these approaches, research suggests that gut microbiota-derived metabolites targeting human G protein-coupled receptors (GPCRs) hold promise for AD treatment (Qiu et al., 2024). The authors developed a machine learning model to explore the relationship between gut microbial metabolites, GPCRs and their potential effect on AD. The model identified orphan GPCRs as potential drug targets to treat AD and the corresponding gut microbial metabolites that interact with these GPCRs. Notably, experimental validation confirmed the effectiveness of this treatment strategy. These studies exemplify the value and effectiveness of applying AI techniques in the drug target discovery process for AD.

4.2 | Multiple sclerosis (MS)

MS is an autoimmune disease where the body's immune system attacks the protective covers of nerve cells in the central nervous system (CNS), including the brain and spinal cord (Dobson & Giovannoni, 2019). AI has been emerging as a powerful tool to aid both diagnosis and treatment development for MS patients. In the realm of diagnosis, AI is proven efficient at analysing medical images. For instance, convolutional neural networks have been trained to identify enhancing lesions with high accuracy on magnetic resonance imaging (MRI) scan data (Narayana et al., 2020). Beyond diagnosis, AI is also contributing to drug discovery for MS. Regression models provide quantitative assessments of a drug's therapeutic potential, estimating its impact on key MS indicators such as inflammation levels and brain lesion reduction. Conversely, classification models offer a binary prediction of a drug's ability to treat MS, categorizing compounds as potentially effective or ineffective. Network-based approaches are being used to conduct drug repositioning studies, where existing drugs are evaluated for potential effectiveness against MS-related genes (deAndrés-Galiana et al., 2019). This method has led to the identification of multiple high discriminatory genes in patients who develop MS and potential relation to infectious agents. Additionally, AI models analysing clinical and imaging data could pinpoint potential new treatments for patients with MS (Falet et al., 2022). For example, anti-CD20 monoclonal antibodies, identified through this approach, show promise in improving long-term survival rates for MS patients (Falet et al. 2022). These studies showcase the value and effectiveness of AI in advancing MS research.

4.3 | CARDIOVASCULAR DISEASE (CVD)

CVD is a general term for any disorder and disease involving the heart or blood vessels and the leading cause of death globally (Gaziano et al., 2006). Novel AI approaches are proving to be useful tools in this fight, aiding researchers in revealing promising drugs and minimizing possible adverse effects. For CVD drug repurposing, both classification and regression models are utilized. Classification models predict interactions between drugs, targets and CVD-related genes, effectively identifying potential candidates for repurposing. Regression models in this context estimate binding affinities between drugs and targets, offering insights into the strength of these interactions. For

FIGURE 3 Future directions for enhanced drug discovery with artificial intelligence (AI). (a) Enhanced data integration: left: streamlining data access and process by integrating different databases, including those focused on drugs, targets, binding interactions, diseases, side effects and patients. This unified platform would facilitate comprehensive data exploration and analysis. Right: constructing multimodal datasets that combine various data modalities such as structure, sequence, image and feature vector data. By leveraging the benefit of each data type, these datasets can provide a more complete picture for AI models to learn from. (b) Advanced AI techniques: - building foundation AI models capable of harnessing distinct data modalities and omics data (genomics, transcriptomics, etc.) for a more comprehensive approach to drug discovery challenges, exploring the potential of quantum AI algorithms to accelerate drug discovery by tackling problems unsolvable with classical computers, and developing efficient and low-resource small AI models tailored to specific drug discovery tasks, which enables broader application across research settings. (c) Effective validation techniques: Implementing more robust validation strategies to rigorously test models and their predictions is crucial in streamlining the drug discovery process.

instance, a network-based approach has been developed to explore the relationship between disease genes and drug targets within the human protein–protein interactome (Cheng et al., 2018). Specifically, this study identified both a drug potentially increasing the risk of coronary artery disease (CAD) and another drug potentially reducing CAD risk. Another network-based approach focuses on repurposing existing drugs for CVD treatment (Paci et al., 2022). Notably, this method incorporates data on adverse drug effects, aiming to decrease potential adverse effects associated with repurposed drugs. Beyond drug discovery, AI models trained on electrocardiogram (ECG) data have shown promise in revealing the patient's response to CVD treatments, paving the way for more personalized and efficient therapeutic strategies (Tison et al., 2022). These studies showcase the value and potential of AI techniques in advancing CVD study.

4.4 | COVID-19

COVID-19, an infectious disease caused by the coronavirus SARS-CoV-2, rapidly evolved into a global pandemic (Ciotti et al., 2020). AI is a critical tool in the fight against COVID-19, particularly in discovering and developing therapeutics (Zhou, Wang, Tang, Nussinov, & Cheng, 2020c). Classification models are employed to predict whether a single drug or a combination of drugs has the potential to effectively treat COVID-19, providing a binary outcome. Complementing this approach, regression models offer more insights by quantifying the inhibition levels between drugs and the spike proteins on SARS-CoV-2. In the early stages of the pandemic, network-based approaches played a significant role. One such study successfully repurposed 16 drugs and 3 drug combinations with the potential to treat COVID-19 (Zhou, Hou, Shen, Huang, et al., 2020a). Based upon this success, researchers have explored other network-based approaches harnessing SARS-CoV-2 virus–host protein–protein interactions or drug–gene–disease associations extracted from knowledge graph (KG) to highlight potential treatments (Zeng, Song, et al., 2020a; Zhou, Hou, Shen, Mehra, et al., 2020b). Notably, **melatonin** has been identified as a potential treatment for reducing the likelihood of a positive COVID-19 test result, while **toremifene** has shown promise in inhibiting the SARS-CoV-2 spike proteins, a critical factor in viral infection (Martin & Cheng, 2020). These studies illustrate the importance of AI in tackling emerging global health threats like COVID-19 by rapidly identifying potential therapeutic options.

5 | CONCLUSION AND PERSPECTIVE

In this review, we examine the efforts made in leveraging AI techniques to facilitate drug discovery, especially in predicting DTIs and its applications to drug development and drug repurposing. We explore the major databases that house essential data for the study of DTI, as well as the toolkits instrumental in processing drugs and target data. Moreover, we study the evaluation techniques vital for assessing the performance of AI models. Subsequently, our investigation delves into

the application of AI models for predicting DTIs with a focus on both classical machine learning and deep learning models. We analyse various model types, including feature-based, similarity-based, sequence-based, image-based and network/graph-based models.

Despite the remarkable progress of the AI methodologies, the AI and biomedical research communities still face formidable challenges. One of the major challenges is the accessibility and quality of the data. AI models require a large amount of high-quality training data during the training procedures; however, biomedical databases often suffer from noise and incompleteness. Additionally, biomedical data require manual annotation from experts, which can be expensive and time-consuming, thus enlarging the gap between labelled and unlabelled data (Sarker et al., 2022). To mitigate this challenge, semi-supervised learning techniques offer a promising solution in leveraging labelled data to extrapolate labels for unlabelled data (Van Engelen & Hoos, 2020). Future solutions may include building a database that integrates information from all three types of databases:- drug-focused database, target-focused database and binding focus database (Figure 3a).

Another significant challenge lies in effectively harnessing available data. Currently, AI models predominantly operate within single modalities, limiting their capacity to leverage diverse data types. Multimodal AI models are increasingly emerging, demonstrating the capability of integrating distinct data sources. There are three commonly used strategies for integrating different data modalities: early, intermediate and late fusion. Late fusion processes data sources separately and then combines outputs. It struggles with weak correlations between sources. Early fusion combines data at the start, capturing correlations well but requiring careful handling of data size. Intermediate fusion finds a middle ground, balancing the benefits of both (Gadzicki et al., 2020). Such models can learn more information for a single given drug and target and thus have better performance than models using single modal, providing a compelling solution to this limitation (Ngiam et al., 2011). For example, one promising solution involves building a biomedical foundation model (Figure 3b). This model integrates various data modalities of both drugs and targets, including image data, SMILES and amino acid sequence data, structure data and fingerprint data. By simultaneously leveraging this rich information, biomedical foundation models have the potential to revolutionize drug discovery by enhancing drug–target understanding and accelerate drug discovery. Additionally, the integration of multi-omics data represents another trend in enhancing the comprehensibility of input data (Subramanian et al., 2020).

Furthermore, effective validation of AI models is still a critical hurdle. AI models in DTI prediction often struggle with generalization, particularly when faced with data distributions that differ from their training sets. This limitation is exacerbated when models are trained on biased or incomplete datasets. To evaluate an AI model's DTI prediction performance, two testing scenarios are employed - hot-start and cold-start testing. Hot-start testing, which assesses the model's ability to predict interactions for known drugs and targets, is relatively less challenging. In contrast, cold-start testing evaluates the model's capacity to predict interactions for novel drugs and targets, presenting a more rigorous challenge to AI models. The cold-start scenario is crucial for assessing a model's generalization ability, as performance often

deteriorates significantly when encountering drugs or targets absent from the training data. This comprehensive evaluation approach helps researchers identify models robust enough to make reliable predictions in real-world drug discovery scenarios, where encountering novel compounds and targets is common. Common practices like random splitting for training and testing datasets can usually lead to overly optimistic AI models, particularly for chemical problems, while scaffold splitting can be too challenging to AI models (Wu et al., 2018). The key solution lies in developing a fairer splitting technique that balances real-world data representation while preventing both overfitting and underfitting. Additionally, to fully validate the effectiveness of AI models, in vitro and in vivo experimental validations and clinical trials for several predicted candidate drugs with high confidence are also crucial (Figure 3c).

Recent strides in large language models (LLMs), such as generative pretrained transformer (GPT) (Radford, 2018) and bidirectional encoder representations from transformers (BERT) (Devlin, 2018), offer promising prospects for advancing drug discovery and DTI prediction by facilitating the integration of heterogeneous data sources (Chakraborty et al., 2023). However, large language models are susceptible to hallucination (Zhang et al., 2023) and biased responses, especially stereotypic and prejudiced responses (Yu et al., 2024), necessitating caution in their application. Small AI models that are trained on small but high-quality datasets can be more accurate and cost-effective compared to large models, and these small AI models can play critical roles in future drug discovery. Quantum computing is another promising frontier in accelerating drug discovery and DTI prediction. While offering incomparable computational speed and the ability to solve previously impossible problems using classical computers (Chauhan et al., 2022), quantum computers are extremely vulnerable to environmental noise, leading to unwanted errors (Resch & Karpuzcu, 2021).

5.1 | Nomenclature of targets and ligands

Key protein targets and ligands in this article are hyperlinked to corresponding entries in the IUPHAR/BPS Guide to PHARMACOLOGY <http://www.guidetopharmacology.org> and are permanently archived in the Concise Guide to PHARMACOLOGY 2023/24 (Alexander et al., 2023).

AUTHOR CONTRIBUTIONS

Y. Yang: Data curation (equal); formal analysis (equal); investigation (equal); methodology (equal); writing—original draft (equal). **F. Cheng:** Conceptualization (equal); funding acquisition (equal); project administration (equal); resources (equal); validation (equal); writing—original draft (equal).

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CONFLICT OF INTEREST STATEMENT

The authors have declared no competing interests.

DATA AVAILABILITY STATEMENT

N/A-Review.

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